INNOVATORS & STARTUPS

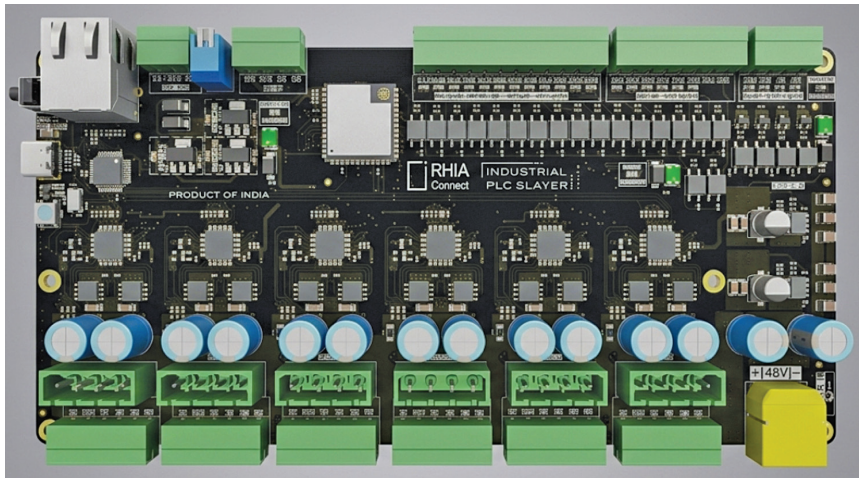
1 RHIA's PLC Slayer Eliminates Fragmented Architectures In Industrial Automation

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1 RHIA'S PLC SLAYER ELIMINATES FRAGMENTED ARCHITECTURES IN INDUSTRIAL AUTOMATION

The startup's innovation integrates separate controllers, sensors, and motor drivers into a single, flexible industrial automation platform.



PLC Slayer integrates industrial control, I/O, motor drivers, and communication interfaces on a compact PCB-based platform

Does building industrial-grade automation always have to be fragmented? Solving a real-world problem while developing custom hardware for Freshot foodbots, Bengaluru-based startup RHIA found itself confronted by this question. The experience made them think differently about hardware architecture, thus sparking the birth of the PLC Slayer, a single, compact platform that unifies the scattered elements.

"We started seeing how industrial machines depended on multiple independent systems for PLC control, motion, motor control, I/O, sensing, and connectivity," says Praveen Olekar, senior business development manager at RHIA. "Bringing all these systems together can make automation complex, expensive, and difficult to integrate."

The Freshot learnings helped shape the creation of a single, unified platform that brings these core functions together. The simplified architecture provides developers, startups, original equipment manufacturers (OEMs), and system integrators with a more flexible starting point for building or modifying automation systems.

At its core, PLC Slayer uses an ESP32 dual-core MCU as its control engine. The platform supports industrial Ethernet, RS485, and controller area network (CAN) bus connectivity, alongside Wi-Fi and Bluetooth. It integrates industrial I/O and six stepper/DC motor drivers, with support for open- and closed-loop motion control and a current capacity of up to 5A per channel.

RHIA is positioning the platform for connected automation through edge-to-cloud connectivity and industrial sensor integration, extending its role beyond standalone machine control.

"The PLC Slayer is designed to be flexible and customisable rather than being limited to a single application. It can be used for applications involving motor control, conveyors, and other industrial automation requirements. We are also seeing applications in areas such as water technology and smart water automation. There is interest from customers working on applications such as vending machines and from small industries," Praveen explains.

The engineering challenges that led to the PLC Slayer also drove RHIA to develop its own industrial sensors and industrial internet of things (IIoT) solutions. This portfolio, covering radio frequency identification (RFID), object detection, and distance measurement, was built entirely to solve specific hardware problems found during customer projects. For example, when commercial sensors failed to accurately measure temperatures between 300°C and 400°C near a cooking area, RHIA developed its